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To the Editor-in-Chief
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Dear Editor-in-Chief,

I am pleased to submit the enclosed manuscript, “Observation Theory: Geometry, Distortion, and Reliability as Properties of Observation,” for consideration as a regular paper in the *IEEE Transactions on Information Theory*.

The paper develops a consumer-relative theory of lossy representation. From a second-order expansion of a downstream task’s distortion it isolates a single controlling quantity—the reconstruction error read through the consumer’s read operator $P_{\mathcal{C}} = J^{\top} G J$ —and shows that ordinary mean-squared-error reconstruction is its isotropic corner $P_{\mathcal{C}} = I$, the corner in which Shannon’s Gaussian rate–distortion function and Gauss least squares reside. The principal contributions are information-theoretic. (i) A consumer-relative rate–distortion function, with achievability and converse for a general source, and a coding theorem for the *true* nonlinear consumer divergence that reduces exactly to output-side coding. (ii) For a Gaussian source read by two operators, the *complete two-stage rate region*: successive refinability is characterized by a single Loewner condition—the observers’ optimal error covariances must nest—with a closed-form max-determinant rate loss when they do not, extended to the k -observer chain. (iii) Two coding-theoretic consequences of the reduction: a finite-blocklength result whose effective dimension is the *rank* of the read operator rather than the ambient dimension, and a source–channel separation theorem for the consumer distortion. (iv) An operational theory of a *misidentified* observer—mis-weighting a read direction costs a bounded rate tax, whereas omitting one erects an irreducible distortion floor that no rate can cross—an asymmetry that, within the Gaussian model, derives a compression catastrophe observed in the experiments. (v) The identification of the read operator with the pullback geometry of *distinguishability*, and its recovery from black-box consumer queries, validated blind against a planted ground truth. Throughout, ordinary reconstruction is the isotropic corner $P_{\mathcal{C}} = I$, and the framework is positioned as the identifiable geometric middle term between Shannon rate–distortion and the information bottleneck.

The work lies squarely within the Transactions’ scope—Shannon theory, rate–distortion and source coding, quantization, and information geometry, with connections to learning and inference. It draws on and extends the indirect/remote source-coding tradition (Dobrushin–Tsybakov, Wolf–Ziv, Witsenhausen), coding for computing (Körner, Orlitsky–Roche), and the rate–distortion–perception and information-bottleneck literatures, all of which are cited and discussed.

The material is original and previously unpublished; it has not been presented, in whole or in part, at any conference, and it is not under review at any other journal. It relates to, but is distinct from, two companion works by the author (on the spectral/angular observer and on geometric key quantization) that are cited in the manuscript; the present paper is a self-contained information-theoretic synthesis and is not a verbatim extension of either.

In the interest of reproducible research, every empirical claim in the paper is governed by a sealed, before-the-fact registration with a pre-declared acceptance bar. The pre-registrations, experiment code, committed result artifacts, and a continuous-integration workflow that re-runs the reproducible experiments and verifies the paper's theoretical constructions are publicly available at the repository named in the manuscript, with a permanent archival snapshot deposited at Zenodo ([doi:10.5281/zenodo.21423316](https://doi.org/10.5281/zenodo.21423316)). In addition, each analytic result was independently re-derived in a fresh-context verification pass whose outcome—including two self-corrections caught before submission—is recorded in the ledger. Consistent with the Society's encouragement, I intend to post a preprint to arXiv at the time of submission.

I confirm that I am the sole author; that the submission is prepared in the required single-column format and is well within the 50-page review limit; and that I hold a registered ORCID. I would be glad to suggest qualified reviewers or to provide any additional material the editors may find helpful.

Thank you for your time and for considering this work.

Sincerely,

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